THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 6, 9, 7, 5, 16, 7 respectively, The mean of the data is: 1

(a) 270 **(b)** 33.4 **(c)** 32.4 **(d)** 35

2. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 10, 6, 15, 12, 5, 2 respectively, The median of the data is: 1

(a) 26 **(b)** 20 **(c)** 25.4 **(d)** 30

3. For the data with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 2, 16, 10, 2, 4, 5, 11 respectively, The mode of the data is: 1

(a) 77 **(b)** 94.8 **(c)** 70 **(d)** 80

4. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 13, 4, p, 8, 14, 5 respectively is 39.2. The value of p is: 1

(a) 5 **(b)** 6 **(c)** 11 **(d)** 8

5. The median of a distribution with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 and frequencies 18, 18, x, 20, y, 12 respectively, and total frequency 100, is 91.5. The value of x is: 1

(a) 23 **(b)** 11 **(c)** 12 **(d)** 10

6. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 21, 3, x, 10, y, 14, 9 respectively, and total frequency 100, is 35. The value of y is: 1

(a) 20 **(b)** 22 **(c)** 23 **(d)** 24

7. If $\sum f_i = 40$ and $\sum f_i x_i = 1200$, the mean is: 1

(a) 30 **(b)** 1160 **(c)** 40 **(d)** 300

8. For the data with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 with frequencies 3, 6, 13, 15, 7, 6 respectively, The mean of the data is: 1

(a) 111 **(b)** 112 **(c)** 113 **(d)** 115

9. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 12, 9, 5, 14, 5, 5 respectively, The median of the data is: 1

(a) 40 **(b)** 36.2 **(c)** 30 **(d)** 38

10. For the data with classes 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 5, 13, 5, 2, 4, 11 respectively, The mode of the data is: 1

(a) 90 **(b)** 80 **(c)** 85 **(d)** 100

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	6	9	7	5	16	7

Find the mean.

12. The marks obtained by students in a test are given below. 2

Marks	20–30	30–40	40–50	50–60	60–70	70–80
Number of students	4	4	10	8	8	6

Find the median.

13. The marks obtained by students in a test are given below. 2

Marks	60–70	70–80	80–90	90–100	100–110	110–120	120–130
Number of students	2	16	10	2	4	5	11

Find the mode.

14. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	80–90	90–100	100–110	110–120	120–130	130–140
Number of workers	3	6	13	15	7	6

Find the mean.

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	10	6	15	12	5	2

Find the mean.

16. The weights (in kg) of students of a class are given below. 3

Weight (kg)	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	6	9	7	5	16	7

Find the median.

17. The marks obtained by students in a test are given below. 3

Marks	40–50	50–60	60–70	70–80	80–90	90–100
Number of students	4	12	10	19	3	2

Find the mean and the mode (in this order).

OR

The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	10–20	20–30	30–40	40–50	50–60	60–70
Number of workers	12	9	5	14	5	5

Find the mean.

18. The daily electricity consumption (in units) of households in a colony is given below. 3

Units consumed	80–90	90–100	100–110	110–120	120–130	130–140
Number of households	3	6	13	15	7	6

Find the median.

OR

The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	10–20	20–30	30–40	40–50	50–60	60–70
Number of workers	13	4	6	8	14	5

Find the mean and the mode (in this order).

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 and frequencies 6, 9, 7, p, 16, 7 respectively is 32.4. Find the value of p. 5

OR

The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 7, 14, x, 20, y, 19, 3 respectively, and total frequency 100, is 54. Find the value of x. 5

20. The mean of a distribution with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 and frequencies 3, 6, p, 15, 7, 6 respectively is 112. Find the value of p. 5

OR

The median of a distribution with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140, 140–150 and frequencies 3, 8, 16, x, 20, y, 10 respectively, and total frequency 100, is 120.5. Find the value of x. 5

END OF QUESTION PAPER Total: 40 Marks